

ROOTPRINT

Kashmir Craft Passport

A voice-first provenance system that turns an artisan's oral knowledge into a QR-verifiable digital identity.

Hackathon Strategy, Pitch, Build Plan, and MVP Specification

1. Executive Summary

One-line pitch

Rootprint gives every Kashmiri artisan a Craft Passport: speak your story once, and the system generates a structured Heritage Record, an authenticity score, and a QR page that any buyer, tourist, professor, or retailer can scan to verify who made the product, where it came from, and why it is real.

The original idea was powerful but too distributed for a hackathon: marketplace, archive, AI agents, tourism, grants, commerce, APIs, dashboards, and institutional licensing all competing for attention. The winning version is simpler:

Do not pitch Rootprint as another marketplace. Pitch it as the missing proof layer for Kashmiri craft.

Rootprint starts with one urgent, locally resonant problem: genuine Kashmiri crafts are losing both **memory** and **market trust**. Artisans cannot prove provenance at the point of sale; buyers cannot distinguish authentic work from machine-made or mislabelled products; and the techniques behind the work remain undocumented.

The hackathon MVP proves the full value chain in one live flow:

1. Record or upload a short artisan interview.
2. Use AI to extract lineage, craft, technique, material, location, and product details.
3. Generate a clean Heritage Record and a transparent confidence/completeness score.
4. Create a QR-linked public Craft Passport page.
5. Show how a buyer scans the QR code and immediately trusts the product.

2. Pain-point Analysis: What Was Weak and How We Fix It

Pain point in original concept	Why a jury may resist it	Fix in the new pitch
Too many products	Seven layers sound like a company roadmap, not a hackathon build.	Compress everything into one MVP: Voice Intake → Heritage Record → QR Craft Passport.
Abstract positioning	“Institutional Memory OS” is impressive but hard to visualise.	Use a concrete phrase: Kashmir Craft Passport . The judge instantly understands identity, proof, and travel of a product.
Marketplace confusion	Judges may ask why this is not Etsy, ONDC, Instagram, or a tourism app.	Make Rootprint the verification and provenance layer , not the marketplace. Marketplaces can plug into it later.
Verification over-claim	Full authenticity verification needs field operations, guilds, and time.	For MVP, provide a completeness and evidence score , with clear fields marked “self-declared”, “document-backed”, or “community-backed”.
Deployment risk	A giant system with agents, commerce, archive, grants, and APIs feels hard to deploy.	Demo a deployable web app: interview upload, AI extraction, record editor, QR generator, public page.
Weak jury hook	The idea is global, but the jury is local to Kashmir.	Anchor emotionally and technically in Kashmir: pashmina, carpets, papier-mache, walnut wood, craft clusters, and NIT Srinagar as the birthplace of the proof standard.
Too much future vision	The strongest idea gets diluted by features that are not needed now.	Present the rest as expansion modules after the core passport works.

The new Rootprint is easier to explain, easier to demo, easier to judge, and still large enough to become a serious company.

3. The USP: The Unfairly Strong Version

Core USP

Rootprint is the first voice-first Craft Passport for Kashmir: it converts an artisan’s oral memory into a scannable provenance identity in minutes.

This is strong because it combines four things that rarely exist together:

1. **Local truth:** Kashmiri craft knowledge is often oral, family-held, and undocumented.
2. **Buyer trust:** Counterfeit and mislabelled “Kashmiri” products damage both artisans and consumers.

3. **Immediate demo:** A judge can scan a QR code and see the entire product story.
4. **Technical depth:** The system uses speech processing, LLM extraction, structured schemas, confidence scoring, multilingual summaries, QR infrastructure, and public verification pages.

3.1. The Sentence to Repeat During the Pitch

“If Kashmir does not create the digital proof standard for Kashmiri craft, someone else will define authenticity for us.”

That sentence makes the project local, urgent, and strategic. It gives the professors a reason to care beyond software.

3.2. Why This Can Win

- **It is not generic AI.** It is AI applied to a real Kashmiri problem with cultural, economic, and technical relevance.
- **It has a live artifact.** The jury can hold a product tag, scan the QR, and see proof.
- **It respects artisans.** The interface begins with voice, not forms or English-first dashboards.
- **It is technically credible.** The MVP is small enough to build, but the architecture can grow into a platform.
- **It has a defensible moat.** The growing corpus of verified Heritage Records becomes the canonical dataset for craft provenance.

4. The Problem, Rewritten for a Jury

Kashmir’s craft economy has three connected failures:

1. **Authenticity failure:** Buyers cannot easily verify whether a pashmina, carpet, papier-mache item, or walnut wood piece is genuinely Kashmiri and handmade.
2. **Memory failure:** Techniques, motifs, material knowledge, family histories, and regional variations live inside artisans’ memories and are rarely structured digitally.
3. **Market power failure:** Middlemen and counterfeit sellers benefit from the lack of proof, while genuine artisans struggle to command premium prices.

The danger is not only that fake products sell. The deeper danger is that authentic knowledge becomes invisible. When a craft has no digital identity, the market treats it like a commodity.

5. The Solution: Rootprint Craft Passport

Rootprint creates a permanent, QR-verifiable digital identity for each artisan, business, or product batch.

5.1. What a Craft Passport Contains

- | | |
|--|------------------|
| • Artisan or business name | • Craft category |
| • Village, district, and craft cluster | • Product type |

- Techniques used
- Material provenance
- Family or apprenticeship lineage
- Process summary
- Photos, audio, or video evidence
- Verification level
- Public QR page
- Last updated timestamp

5.2. The Key Design Choice

Most systems ask artisans to fill forms. Rootprint starts with conversation.

Voice-first advantage

An artisan should be able to speak naturally in Kashmiri, Urdu, Hindi, or English. The system does the technical work: transcription, translation, extraction, structuring, and presentation.

6. MVP Demo Flow

The demo must be simple enough that the jury can describe it back in one sentence.

Step	What the jury sees
1. Speak	A short artisan interview is recorded or uploaded.
2. Extract	AI pulls out craft, technique, material, origin, lineage, and product claims.
3. Review	The team shows a structured Heritage Record with editable fields.
4. Score	Rootprint computes a transparent Passport Completeness Score.
5. Publish	A QR code is generated for the artisan/product.
6. Verify	The judge scans the QR and opens a clean mobile provenance page.

Live demo script

“This is not a marketplace listing. This is a proof document. In two minutes, we convert a spoken artisan story into a digital passport that can travel with the product anywhere in the world.”

7. Technical Architecture

7.1. MVP System Components

Component	Responsibility
Voice intake	Record or upload artisan audio/video; store raw media reference.
Speech-to-text	Convert speech to transcript; prototype can use a hosted API or local model depending on available time.
LLM extraction	Convert transcript into structured JSON following the Heritage Record schema.
Record editor	Human review screen to correct fields before publishing.
Scoring engine	Calculate Passport Completeness Score based on filled fields and evidence type.
QR generator	Create unique QR code for each public passport URL.
Public passport page	Mobile-first page showing story, craft, evidence, and verification status.
Admin dashboard	List passports, view scores, update records, and copy QR links.

7.2. Suggested Heritage Record Schema

```
{
  "artisan": {
    "name": "",
    "location": "",
    "languages": [],
    "generation": ""
  },
  "craft": {
    "category": "",
    "techniques": [],
    "tools": [],
    "materials": []
  },
  "provenance": {
    "source_region": "",
    "lineage_story": "",
    "evidence": []
  },
  "product": {
    "name": "",
    "description": "",
    "price_range": ""
  },
  "verification": {
    "status": "self_declared | document_backed | community_backed",
    "score": 0,
    "last_updated": ""
  }
}
```

7.3. Scoring That Avoids Overclaiming

The MVP should not pretend to legally certify authenticity. Instead, it should score evidence quality.

Evidence dimension	Points
Artisan identity and location provided	15
Craft category and technique specificity	20
Material source or supplier detail	15
Lineage, apprenticeship, or family history	15
Media evidence uploaded	15
Human review completed	10
Community or document reference attached	10
Total	100

Use **Passport Completeness Score** in the prototype. Reserve **Certified Authentic** for later when guild, government, or expert verification partnerships exist.

8. Product Screens to Build

1. **Landing page:** “Give your craft a passport.” One button: Create Passport.
2. **Interview page:** Record/upload audio, choose language, add optional product photo.
3. **AI extraction review:** Transcript on left, extracted fields on right, confidence indicators.
4. **Passport dashboard:** List of created passports with score, status, QR, and public link.
5. **Public QR page:** Buyer-facing mobile page with artisan story, craft facts, evidence, and scan timestamp.

8.1. The Public QR Page Must Look Premium

The buyer-facing page is the emotional center of the demo. It should not feel like a database row. It should feel like a museum label, certificate, and product story combined.

- Hero: artisan name, craft, location, verification level.
- “Why this is real”: technique, material, lineage, evidence.
- “Listen to the maker”: short audio clip or transcript excerpt.
- “About this craft”: short educational section for tourists/buyers.
- “Passport ID”: immutable-looking ID, timestamp, QR.

9. What to Remove from the Hackathon Pitch

These ideas are still valuable, but they should be moved to “future roadmap” so the pitch does not look impossible:

- Full direct commerce marketplace
- Seven autonomous AI agents
- Grant application automation

- Tourism marketplace and live workshop bookings
- Enterprise API licensing
- Large-scale government data contracts
- Legal certification claims

How to mention the larger vision

“Today we are building the passport. Tomorrow, commerce, tourism, grants, and institutional APIs can plug into that passport. The passport is the foundation.”

10. Roadmap: From Hackathon to Real Deployment

Stage	Scope
Hackathon MVP	One craft category, 3–5 sample passports, AI extraction, scoring, QR pages.
Pilot	25–50 real artisans, human-reviewed records, local language interviews, printable product tags.
Verification layer	Partnerships with craft experts, guilds, departments, or academic reviewers for community-backed records.
Buyer layer	Scan analytics, product inquiry button, export-ready PDF certificate.
Platform layer	APIs for retailers, tourism boards, institutions, and marketplaces.

11. Team Division: 5 People as 2 + 2 + 1

Unit	Ownership	Deliverables
Team A: 2 people	Frontend and demo experience	Landing page, interview/upload screen, extraction review UI, dashboard, public QR passport page, premium visual polish.
Team B: 2 people	Backend and data infrastructure	API routes, database schema, media storage, QR generation, passport URLs, scoring engine, deployment, seed data.
Solo track: 1 person	AI pipeline, schema, pitch integration, and final demo orchestration	Prompt design, transcript-to-JSON extraction, confidence handling, multilingual story generation, Heritage Record schema, demo script, sample artisan data, final PDF/pitch alignment.

11.1. Parallel Work Plan

- First integration target:** Backend exposes create/read passport endpoints; frontend can create a dummy passport; AI pipeline can produce valid JSON from a sample transcript.
- Second integration target:** Real transcript creates a passport record; QR opens the public page.
- Final integration target:** Demo mode with one polished sample pashmina or carpet passport, backup seed data, and a rehearsed scan flow.

If live speech recognition fails, use a prerecorded artisan-style interview transcript. The core demo is not the microphone; the core demo is conversion of oral knowledge into a passport.

12. Pitch Flow for the Final Presentation

1. **Opening hook:** “Everyone knows Kashmiri craft. Very few buyers can prove it.”
2. **Problem:** Authentic artisans lose value because memory and provenance are not digitally portable.
3. **Demo promise:** “We will create a Craft Passport live.”
4. **Demo:** Record/extract/review/score/publish/scan.
5. **Technology:** Voice-first AI extraction, structured schema, transparent scoring, QR provenance.
6. **Why Kashmir first:** Global recognition, counterfeit pressure, deep craft diversity, local institutional relevance.
7. **Business model:** Subscription for artisans/businesses, QR tags, verification review, export certificates, API integrations later.
8. **Impact:** Higher trust, better pricing power, preserved knowledge, local ownership of authenticity data.
9. **Closing line:** “Rootprint makes Kashmiri craft impossible to erase and harder to fake.”

12.1. 60-second Version

Kashmiri craft has a proof problem. A genuine artisan may spend years mastering a technique, but when the product reaches a buyer, it competes with fakes that use the same words: handmade, authentic, Kashmiri. Rootprint solves this with a voice-first Craft Passport. The artisan speaks their story; AI turns it into a structured Heritage Record; the system generates a QR page that travels with the product. A buyer scans and sees who made it, where it came from, what technique was used, and what evidence supports the claim. We are not building another marketplace. We are building the proof layer that marketplaces, tourism, export, and preservation can trust.

13. Business Model, Kept Simple

Revenue stream	Description
Passport creation	Free basic passports; paid upgraded passports with more media, printable QR tags, and analytics.
Verification review	Fee for expert/community-backed review once partnerships exist.
QR tag packs	Printed durable tags for products, workshops, stores, and export packaging.
Export certificate PDFs	Paid generation of buyer-ready provenance and material documentation.
Retail/API integrations	Later-stage integrations for stores that want verified craft data embedded in product pages.

14. Judging Criteria Alignment

Likely judging criterion	Rootprint answer
Innovation	Voice-first AI provenance passport for living craft knowledge.
Technical implementation	Speech-to-text, LLM extraction, JSON schema, scoring, QR generation, public verification pages.
Social impact	Helps artisans preserve knowledge, build trust, and improve pricing power.
Local relevance	Starts with Kashmir's globally recognised craft identity.
Feasibility	Narrow MVP with clear demo flow and realistic team split.
Scalability	Same passport model can extend to other crafts and regions after Kashmir.

15. Final Positioning

What Rootprint is

Rootprint is a **digital proof standard** for handmade heritage products.

What Rootprint is not

Rootprint is not trying to replace artisans, become another e-commerce marketplace, or make legal authenticity claims without community verification.

Closing line

Rootprint makes Kashmiri craft impossible to erase and harder to fake.